

Brandon Williams

CONTACT INFORMATION	Heidelberg University Institute of Mathematics Im Neuenheimer Feld 205 69120 Heidelberg	<i>E-mail:</i> bwilliams@mathi.uni-heidelberg.de <i>WWW:</i> btw-47.github.io
RESEARCH INTERESTS	Modular forms and their generalizations, including vector-valued modular forms, Jacobi forms, mock modular forms and automorphic forms on more general groups.	
EDUCATION	University of California, Berkeley, CA	
	Ph.D., Mathematics	2013-2018
	• Dissertation title: <i>Computing modular forms for the Weil representation</i> • Adviser: Prof. Richard Borcherds • Area of Study: Automorphic forms	
	Heidelberg University, Germany	
	M.Sc., Mathematics	2011-2013
	• Thesis title: <i>On elliptic curves with complex multiplication, L-functions, and p-adic interpolation</i> • Adviser: Prof. Dr. Otmar Venjakob	
	B.Sc., Mathematics	2009-2011
	• Thesis title: <i>GARCH(1,1)-models</i> • Adviser: Prof. Dr. Rainer Dahlhaus	
EMPLOYMENT	Heidelberg University, Germany	
	Postdoctoral researcher	2024-Present
	• Postdoc mentor: Prof. Dr. Georg Oberdieck	
	RWTH Aachen, Germany	
	Postdoctoral researcher	2020-2024
	• Postdoc mentor: Prof. Dr. Aloys Krieg	
	Technische Universität Darmstadt, Germany	
	Postdoctoral researcher	2018-2020
	• Postdoc mentor: Prof. Dr. Jan Bruinier	
AWARDS	• 2018 Ken Ribet – Lisa Goldberg Award in Algebra (department dissertation award) • Lehmer Fellowship in Number Theory, Spring 2018.	

PUBLICATIONS

- [1] Williams, B. *Poincaré square series for the Weil representation*. Ramanujan J. 47 (2018), no. 3, 605–650.
- [2] Williams, B. *Vector-valued Hirzebruch–Zagier series and class number sums*. Res. Math. Sci. 5 (2018), no. 2, Paper No. 25, 13 pp.
- [3] Williams, B. *Rankin–Cohen brackets and Serre derivatives as Poincaré series*. Res. Number Theory 4 (2018), no. 4, Art. 37, 13 pp.
- [4] Williams, B. *Vector-valued Eisenstein series of small weight*, Int. J. Number Theory 15 (2019), no. 2, 265–287.
- [5] Williams, B. *Poincaré square series of small weight*. Ramanujan J. 48 (2019), no. 3, 585–612.
- [6] Williams, B. *Overpartition M_2 -rank differences, class number relations, and vector-valued mock Eisenstein series*. Acta Arith. 189 (2019), no. 4, 347–365.
- [7] Pribitkin, W. and Williams, B. *Short proof of Rademacher’s formula for partitions*. Res. Number Theory 5 (2019), no. 2, Art. 17, 6 pp.
- [8] Williams, B. *Remarks on the theta decomposition of vector-valued Jacobi forms*, J. Number Theory 197 (2019), 250–267.
- [9] Schwagenscheidt, M. and Williams, B. *Twisted component sums of vector-valued modular forms*, Abh. Math. Sem. Univ. Hamburg 89 (2019), no. 2, 151–168.
- [10] Williams, B. *A construction of antisymmetric modular forms for Weil representations*, Math. Zeitschrift 296 (2020) 391–408.
- [11] Williams, B. *Graded rings of paramodular forms of level 5 and 7*, J. Number Theory 209 (2020) 483–515.
- [12] Williams, B. *The rings of Hilbert modular forms for $Q(\sqrt{29})$ and $Q(\sqrt{37})$* , J. Algebra 559 (2020) 679–711.
- [13] Wang, H. and Williams, B. *On some free algebras of orthogonal modular forms*, Adv. Math. 373 (2020), Article 107332.
- [14] Williams, B. *Higher pullbacks of modular forms on orthogonal groups*, Forum Math. 33 (2021), 631–652.
- [15] Wang, H. and Williams, B. *Borcherds products of half-integral weight*, J. Number Theory 238 (2022), 944–950.
- [16] Williams, B. *Two graded rings of Hermitian modular forms*, Abh. Math. Sem. Univ. Hamburg 91 (2021) 257–285.
- [17] Wang, H. and Williams, B. *Graded rings of Hermitian modular forms with singularities*, Manuscripta Math. 170 (2023) 283–311.
- [18] Wang, H. and Williams, B. *Siegel modular forms of degree two and level five*, Ramanujan J. 60 (2023) 597–613.
- [19] Hauffe-Waschbüsch, A. and Krieg, A. and Williams, B. *On Hermitian Eisenstein series of degree 2*, Funct. Approx. 68(1) (2023), 127–141.
- [20] Wang, H. and Williams, B. *Projective spaces as orthogonal modular varieties*, Transform. Groups 30 (2025) 933–966.
- [21] Wang, H. and Williams, B. *Simple lattices and free algebras of modular forms*, Adv. Math. 413 (2023), Article 108835.

- [22] Schwagenscheidt, M. and Williams, B. *Binary theta functions and Borcherds products*, J. Number Theory 249 (2023) 441–461.
- [23] Wang, H. and Williams, B. *On weak Jacobi forms of rank two*, J. Algebra 634 (2023) 722–754.
- [24] Wang, H. and Williams, B. *Modular forms with poles on hyperplane arrangements*, Algebraic Geom. 11(4) (2024) 506–568.
- [25] Wang, H. and Williams, B. *Automorphic products that are singular modulo primes*, Res. Number Theory 10 (2024), article 8.
- [26] Wang, H. and Williams, B. *The fake monster algebra and singular Borcherds products*, Adv. Math. 461 (2025), 110083.
- [27] Wang, H. and Williams, B. *Mathieu moonshine and Borcherds products*, Commun. Number Theory Phys. 19(1) (2025) 135–168.
- [28] Wang, H. and Williams, B. *Free algebras of modular forms on ball quotients*, Res. Math. Sci. 12, art. 61 (2025)
- [29] Clingher, A. and Malmendier, A. and Williams, B. *K3 surfaces and orthogonal modular forms*, Nagoya Math. J. 260 (2025) 687–727.
- [30] Sun, K. and Wang, H. and Williams, B. *Hyperbolization of affine Lie algebras*, Mem. Amer. Math. Soc., to appear.
- [31] Wang, H. and Williams, B. *On the non-existence of singular Borcherds products*, Amer. J. Math., to appear.
- [32] Barros, I. and Beri, P. and Flapan, L. and Williams, B. *Cones of Noether–Lefschetz divisors and moduli spaces of hyperkähler manifolds*, Math. Ann., to appear.
- [33] Oberdieck, G. and Williams, B. *Quasi-modular forms for the orthogonal group and Gromov–Witten theory of Enriques surfaces*, Commun. Number Theory Phys., to appear.
- [34] Ibukiyama, T. and Williams, B. *Hermitian modular forms and algebraic modular forms on $SO(6)$* , J. Number Theory 286 (2026) 230–279.
- [35] Clingher, A. and Malmendier, A. and Williams, B. *On the Vinberg family of K3 surfaces*, 2026, arXiv: 2606.07849

SUBMITTED
PREPRINTS

CONFERENCES AND
SEMINARS

- *Poincaré square series for the Weil representation* in Trends in modular forms, Daejeon, South Korea, December 19–22, 2017.
- *Computing obstruction spaces for Borcherds products* in ABKLS seminar, Aachen, Germany, February 14, 2018.
- *Vector-valued Hirzebruch–Zagier series and class number sums* in Emory University Algebra and Number Theory Seminar, Atlanta, GA, April 17, 2018.
- *Class number sums* in Universität Köln Oberseminar Zahlentheorie und Modulformen, Cologne, Germany, June 12, 2018.
- *Hilbert modular forms and Borcherds products* in Chalmers University of Technology, Algebraic Geometry and Number Theory Seminar, Gothenburg, Sweden, November 21, 2018.
- *Hilbert modular forms and Borcherds products* in Darmstadt Algebra Group Winter Seminar, Manigod, France, February 22, 2019.

- *Hilbert modular forms and Borcherds products* in Dartmouth College Number Theory Seminar, Hanover, NH, May 23, 2019.
- *Higher pullbacks of modular forms on orthogonal groups* in Darmstadt Algebra Seminar, Darmstadt, Germany, February 18, 2020.
- *Higher pullbacks of modular forms on orthogonal groups in Integrable systems and automorphic forms*, Sochi, Russia, February 23-27, 2020.
- *Free algebras of orthogonal modular forms* in Darmstadt Algebra Group Winter Seminar, Manigod, France, March 13, 2020.
- *Simple lattices and free algebras of modular forms*, RWTH Aachen, January 18, 2021. (Online)
- *Simple lattices and free algebras of modular forms*, Brown University, March 1, 2021. (Online)
- *Borcherds products and a ring of Hermitian modular forms*, University of Oregon, May 3, 2021. (Online)
- *Weak Jacobi forms of rank two*, 34th Automorphic Forms Workshop, March 19, 2022. (Online)
- *Free algebras of modular forms on complex ball quotients* in Universität Bielefeld Oberseminar Gruppen und Geometrie, Bielefeld, Germany, April 27, 2022.
- *Free algebras of modular forms on complex ball quotients* in International Seminar on Automorphic Forms, May 3, 2022. (Online)
- *Additive theta lifts that are Borcherds products* in ENTR Workshop, Darmstadt, Germany, October 27, 2022.
- *Modular products that are singular modulo primes*, KTH Royal Institute of Technology, Stockholm, Sweden, June, 7, 2023.
- *Computation of vector-valued modular forms* (Lightning talk) in *LuCaNT*, ICERM, Providence, RI, July 13, 2023.
- *Computing Fourier coefficients of paramodular eigenforms*, ETH Zürich, Switzerland, October 20, 2023.
- *Modular forms with poles on hyperplane arrangements*, Research on Automorphic Forms, RIMS, Kyoto, Japan, January 24, 2024.
- *Modular forms that are singular modulo primes*, Seminar, Osaka University, Osaka, Japan, February 2, 2024.
- *Modular forms with poles on hyperplane arrangements*, Algebraic Geometry seminar, Sapienza University, Rome, Italy, February 28, 2024.
- *Cones of Noether–Lefschetz divisors and moduli spaces of hyperkähler manifolds*, Heidelberg Algebraic Geometry Seminar, Heidelberg, Germany, January 24, 2025.
- *Hermitian modular forms and orthogonal modular forms*, Automorphic Forms in Budapest 2024, Budapest, Hungary, August 30, 2024.
- *Hermitian modular forms and orthogonal modular forms*, Modular in Bielefeld, Bielefeld, Germany, June 23, 2025.
- *Quasi-modular forms for the orthogonal group*, International Conference on Modular Forms, Wuhan, China, July 8, 2025.

- *K3 surfaces and modular forms*, International Workshop on Automorphic Forms, SwissMAP Research Station, Les Diablerets, Switzerland, September 5, 2025.
- *K3 surfaces and modular forms*, Algebraic Geometry Seminar, Heidelberg, Germany, December 5, 2025.
- *Two families of K3 surfaces associated to root lattices*, Algebraic Geometry Seminar, Universiteit Antwerpen, Belgium, March 20, 2026.
- *Two families of K3 surfaces associated to root lattices*, Amplitudes and Algebraic Geometry, ESI, Vienna, Austria, March 24, 2026.

TEACHING
EXPERIENCE

Heidelberg University, Germany

Instructor

2024-present

- Jacobi forms (Special lecture)
- Quadratic forms (Special lecture)
- Funktionentheorie I (Complex analysis I)

Teaching assistant

2024-present

- Representation theory

RWTH Aachen University, Germany

Instructor

2020-2024

- Complex analysis II
- Analytic number theory

Teaching assistant

2020-2024

- Smart Study Start (Introductory mathematics course)

Student seminars

2020-2024

- Modular forms
- Elliptic curves

Technische Universität Darmstadt, Germany

Teaching assistant

2019-2020

- Analysis II (English)
- Linear Algebra I (English)

University of California, Berkeley, CA

Instructor (GSI)

Summers 2014-2016

- Math 16B (Analytic geometry and calculus)
- Math 54 (Linear algebra and differential equations)
- Math 110 (Linear algebra)

Teaching assistant (GSI)

2013-2017

- Math 16A (Analytic geometry and calculus)
- Math 53 (Multivariable calculus)
- Math 54 (Linear algebra and differential equations)
- Math 55 (Discrete mathematics)
- Math H54 (Honors linear algebra and differential equations)
- Math 110 (Linear algebra)
- Math 185 (Introduction to complex analysis)
- Math 250A (Groups, rings, and fields)

Ruprecht-Karls-Universität Heidelberg, Germany

Teaching assistant (Tutor)

2011-2013

- Linear algebra
- Algebra 1 (field and Galois theory)
- Elementary number theory

**OTHER
EXPERIENCE**

Ruprecht-Karls-Universität Heidelberg, Germany

- **Data mining practicum** (Summer 2012) Analyzed clusters in graphs of OSM geographic data using a variant of the PageRank algorithm.

Heidelberg Collaboratory for Image Processing, Heidelberg, Germany

- **Computer vision practicum** (Summer 2011) Used histograms of oriented gradients and related image descriptors to analyze illuminated manuscripts.

SOFTWARE

Programming languages:

- I am proficient with Python.
- I have some experience programming in C++ and Julia.

Mathematical software:

- Mathematica, MATLAB, SageMath

Other:

- $\text{\TeX}$ ($\text{\LaTeX}$, $\text{\BibTeX}$),
- Microsoft Office, LibreOffice, Google Docs

LANGUAGES

English, German (fluent)
Dutch, French, Russian (intermediate)